A REPRESENTABLE MORPHISM OF ALGEBRAIC SPACES IS FINITE LOCALLY FREE EXACTLY WHEN IT IS AFFINE WITH FINITE LOCALLY FREE PUSHFORWARD OF THE STRUCTURE SHEAF

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement refers to the representable sense of finite local freeness fixed in Section section-representable of the same chapter; the section label is a navigation pointer and the property is spelled out again in the first line of the proof.

Editorial note. The seven references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- The base change formula for pushforwards of modules along an etale morphism. Properties of Spaces, Lemma spaces-properties-lemma-pushforward-etale-base-change-modules, used at proof lines 9581-9584 and 9593-9596. Identifies the restriction of f_*O_X to V with the pushforward along the base change.
- Finite local freeness of a module is local on the etale site. Modules on Sites, Lemma sites-modules-lemma-local-final-object, used at proof lines 9585-9588. Concludes finite local freeness of f_*O_X from its restriction to a cover.
- Effective etale descent for morphisms with a sheaf-theoretic property. Spaces, Lemma spaces-lemma-morphism-sheaves-with-P-effective-descent-etale, with Morphisms, Lemma morphisms-lemma-base-change-finite-locally-free and Descent, Lemmas descent-lemma-descending-property-finite-locally-free and descent-lemma-affine, used at proof lines 9598-9604. Descends finite local freeness and affineness from the etale cover back to f.
- The definition of finite local freeness for representable morphisms. Section section-representable, used at proof lines 9567 and 9573-9578. Fixes the meaning of the property being characterized.

Source reference. Original lemma 0245: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2766–2766. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Date: stacks-project:spaces-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 02VO: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3775–3775. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 02KD: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3368–3368. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original section 03HA: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4559–4559. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 03DN: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4426–4426. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 03I2: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4588–4588. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 03LX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4729–4729. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let S be a scheme. Let $f : X \to Y$ be a representable morphism of algebraic spaces over S. Then f is finite locally free (in the sense of Section 03HA) if and only if f is affine and the sheaf $f_*\mathcal{O}_X$ is a finite locally free $\mathcal{O}_Y$-module.*

Proof. Assume f is finite locally free (as defined in Section 03HA). This means that for every morphism $V \to Y$ whose source is a scheme the base change $f' : V \times_Y X \to V$ is a finite locally free morphism of schemes. This in turn means (by the definition of a finite locally free morphism of schemes) that $f'_*\mathcal{O}_{V \times_Y X}$ is a finite locally free $\mathcal{O}_V$-module. We may choose $V \to Y$ to be surjective and étale. By Properties of Spaces, Lemma 03LX we conclude the restriction of $f_*\mathcal{O}_X$ to V is finite locally free. Hence by Modules on Sites, Lemma 03DN applied to the sheaf $f_*\mathcal{O}_X$ on $Y_{spaces,étale}$ we conclude that $f_*\mathcal{O}_X$ is finite locally free.

Conversely, assume f is affine and that $f_*\mathcal{O}_X$ is a finite locally free $\mathcal{O}_Y$-module. Let V be a scheme, and let $V \to Y$ be a surjective étale morphism. Again by Properties of Spaces, Lemma 03LX we see that $f'_*\mathcal{O}_{V \times_Y X}$ is finite locally free. Hence $f' : V \times_Y X \to V$ is finite locally free (as it is also affine). By Spaces, Lemma 03I2 we conclude that f is finite locally free (use Morphisms, Lemma 02KD Descent, Lemmas 02VO and 0245). Thus we win. □